nature portfolio

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Last updated by author(s): May 23, 2024

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our Editorial Policies and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed	
☐	☒	The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☒	☐	A description of all covariates tested
☒	☐	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒	For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	IVIS Lumina system (Perkin Elmer), LSRFortessa X-50 (BD Biosciences), Beckman Coulter flow cytometer (BD Biosciences), LSM 780 confocal microscope (Zeiss), plate reader (Agilent BioTek), Zetasizer Nano (Malvern), ChemiDoc Imaging System (Bio-Rad Laboratories).
Data analysis	Graphpad Prism v9, FlowJo v10, Living Image Software (Perkin Elmer) v4, Image Lab v6, Microsoft Excel v16. The quality of RNA-Seq reads was assessed with FastQC v0.11.9. Reads from individual samples were aligned using STAR aligner version 2.7.6a to the mouse reference genome GRCm39. The differential gene-expression analysis was performed with Bioconductor package DESeq2 v1.30.0 using the normalized and filtered counts per gene from the RNA-Seq data. The visualization of upregulated genes and associated gene-ontology functions when comparing the three biological conditions was carried out using the Triwise R package.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

The data supporting the results in this study are available within the paper and its Supplementary Information. Source data for the figures are provided with this paper and in figshare at https://doi.org/10.6084/m9.figshare.25919746. The raw and analysed datasets generated during the study are available for research purposes from the corresponding authors on reasonable request.

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	The study did not involve human research participants.
Reporting on race, ethnicity, or other socially relevant groupings	–
Population characteristics	–
Recruitment	–
Ethics oversight	–

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	No power analysis on sample size was performed. In vivo studies on biodistribution were conducted with n>=3; on monitoring T-cell responses were conducted with n=5; on monitoring tumour growth were conducted with n>= 6 tumour-bearing mice per condition. Differences between treatment groups in terms of average T-cell responses or tumour size at the endpoint of the experiment were analysed using one-way ANOVA and Tukey's multiple comparisons test with alpha set to 0.05. All cell samples were evaluated in at least three biological triplicates to allow for statistical analyses. All RNA-stability experiments were performed with three independent replicates. All other sample-size numbers are listed in the corresponding figure captions. Sample sizes for these experiments were chosen on the basis of standards in the field and on prior knowledge of experimental variation.
Data exclusions	No data were excluded.
Replication	All attempts at replication were successful (at least twice at different times).
Randomization	For the in vivo animal study, animal groups were randomized while body weights (20–25g) were maintained within the selected ranges. For the other experiments, all samples were randomly assigned into experimental groups.
Blinding	Blinding was not relevant to the study because the results are purely based on objective measurements.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☐	☒ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used

Antibodies used in this study are: CD45 Brilliant Violet 421 30-F11 BioLegend 103133;CD11c Alexa Fluor 594 N418 BioLegend 117346; CD11b FITC M1/70 BioLegend 101205;CD8a APC/Cy7 53-6.7 BioLegend 100713;CD4 PerCP/Cy5.5 GK1.5 BioLegend 100433;CD103 Brilliant Violet 605 2E7 BioLegend 121433;CD205 PE/Cy7 NLDC-145 BioLegend 138209;F4/80 APC/Cy7 BM8 BioLegend 123117;CD11b PE/Cy5 M1/70 BioLegend 101209;NK1.1 APC S17016D BioLegend 156505;CD3 PerCP/Cy5.5 17A2 BioLegend 100217;CD279(PD-1) Brilliant Violet 421 29F.1A12 BioLegend 135217;IFN-γ PE XMG1.2 BioLegend 505807;TNF-α FITC MP6-XT22 BioLegend 506303;CD44 PE/Cy5 IM7 BioLegend 103009;CD62L FITC MEL-14 BioLegend 104405;CD25 FITC 3C7 BioLegend 101907;FoxP3 Alexa Fluor 647 MF-14 BioLegend 126407;CD279(PD-1) N/A RMP1-14-CP162 Bio X Cell CP162;CD152(CTLA-4) N/A 9H10-CP146 Bio X Cell CP146;CD4 N/A YTS 177 Bio X Cell BP0003-3;CD8α N/A YTS 169.4 Bio X Cell BP0117;NK1.1 N/A PK136 Bio X Cell BP0036;Phospho-PKR (Thr446) N/A Thermo Fisher PA5-121356;EIF2AK2/PKR N/A Thermo Fisher 18244-1-AP;FLAG M2 N/A Sigma-Aldrich F1804;β-actin N/A 15G5A11/E2 Thermo Fisher MA1-140.

Validation

All primary antibodies were bought from vendors (Biolegend, Bio X Cell, Sigma-Aldrich, and Thermo Fisher), and were used for the species suggested by the manufacturers (mouse-specific). All antibodies were validated by the supplier. Validation statements are provided on the manufacturer's website.

Eukaryotic cell lines

Policy information about cell lines and Sex and Gender in Research

Cell line source(s)

The EG7.OVA, TC-1, DC2.4, MC38, B16F10 and HEK293T cell lines were purchased form ATCC. PRR reporter cells were provided by the laboratories of James Moon and Connie Wu (University of Michigan). B3Z cells were originally from Xiaoyuan Chen's lab. The BrafV600E SM1 cell line was from Anthony Ribas' lab at the University of California Los Angeles, USA.

Authentication

The cell lines were authenticated by morphology.

Mycoplasma contamination

The cell lines tested negative for mycoplasma contamination.

Commonly misidentified lines
(See ICLAC register)

No commonly misidentified cell lines were used.

Animals and other research organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research, and Sex and Gender in Research

Laboratory animals

Unless denoted otherwise, 6–8-week old female C57BL/6 mice were used. 12–18 month old female C57BL/6 mice were used in part of the studies of T-cell responses elicited by circRNA vaccines. Mice were given free access to food and water, and were maintained under a 12/12-h light–dark cycle with controlled temperature (20-25°C) and humidity (50 ± 10%).

Wild animals

The study did not involve wild animals.

Reporting on sex

We did not contemplate sex-based analyses.

Field-collected samples

The study did not involve samples collected from the field.

Ethics oversight

All animal work was conducted in compliance with the Virginia Commonwealth University and the University of Michigan Guide for the Care and Use of Animals under protocols approved by the Virginia Commonwealth University Animal Care and by the University of Michigan Animal Care and Use Committee.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Flow Cytometry

Plots

Confirm that:

☒ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).

☒ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).

☒ All plots are contour plots with outliers or pseudocolor plots.

☒ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation	For tetramer staining, peripheral blood was collected from the vaccinated mice, and blood cells were enriched by centrifugation. Red blood cells were lysed using ACK lysis buffer for 10 min at room temperature. Blood clots were removed using a filter. Cells were washed twice in PBS and then stained using Zombie Aqua Fixable Viability Kit (BioLegend) for 20 min at room temperature. Staining was quenched and cells were washed with FCS buffer (PBS buffer with 0.1% FBS). Cells were then blocked with αCD16/CD32 (BioLegend) for 10 min, followed by adding staining cocktail (PerCP/Cy5.5 anti-mouse CD3, APC/Cy7 anti-mouse CD8a, Brilliant Violet 421 anti-mouse PD-1, and tetramer-PE) and staining at room temperature for 30 min. Cells were then washed and 100 µL Cytofix was added into each well to resuspend cells prior to fixation at 4 °C for 20 min. Cells were then washed with Perm/Wash buffer, and resuspended for flow cytometric analysis. For intracellular cytokine staining, Peripheral blood was collected from immunized mice. Red blood cells were removed using ACK lysis buffer, and the obtained lymphocytes were transferred into U-bottom 96-well plates in 200 µL T cell culture media (RPMI 1640, 10% FBS, 100 U/mL penn/strep, 50 µM β-mercaptoethanol, 1x MEM non-essential amino acid solution, and 1 mM sodium pyruvate). Lymphocytes were pulsed with antigen peptides (40 µg/mL) for 4 h, followed by addition of GolgePlug Protein Transport Inhibitor containing brefeldin A (Thermo Fisher Scientific). The sequences of antigen peptides (CSBio, Menlo Park, CA) are shown in Supplementary Table 4. Cells were then placed in a culture incubator for 6 h before incubation with αCD16/CD32 for 10 min at room temperature. Cells were stained with APC/Cy7 anti-mouse CD8a, PerCP/Cy5.5 anti-mouse CD4 and Zombie Aqua Fixable Viability Kit for 20 min at room temperature. Cells were washed and subsequently fixed using Cytofix (BD Biosciences), washed, and permeabilized in 200 µL Cytoperm solution (BD Biosciences). Cells were then washed using Perm/Wash Buffer (BD Biosciences), and permeabilized cells were then stained using PE anti-mouse IFN-γ (BioLegend) and FITC anti-mouse TNF-α (BioLegend). Stained cells were washed for flow cytometric analysis.
Instrument	LSRFortessa X-50 (BD Biosciences), Beckman Coulter flow cytometer (BD Biosciences).
Software	FlowJo V10
Cell population abundance	Cell populations of interest were abundant.
Gating strategy	For antigen presentation, co-stimulatory factors and cell uptake, the cells were first gated based on FSC/SSC and FSCA/FSCH to select for live single cells. Unedited cells were employed as negative control for gating the positive signal. For tetramer staining on T cells, see Supplementary Fig. 32. The cells were first gated based on FSCA/FSCH to select for single cells, then gated based on FSCA/SSCA to select for lymphocytes. Then cells were gated based on SSCA/ Live/Dead to select for live cells. The live cells were gated based on SSCA/CD8 to select for CD8 positive cells. These cells were gated based on CD8/tetramer to select for epitope-positive CD8-positive cells.

☒ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.